MATH 345 Skeleton Notes

Unit 2: Anatomy of a linear map

Section 2.2: Linear systems

The preview examples had different stories: reachable outputs, invisible changes, redundant features. Algebraically, they all ask the same question: can a list of linear equations in the unknown input coordinates be solved?

Definitions and examples

Definition: Linear equations An equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where $a_1, \dots, a_n, b$ are scalars, and $x_1, \dots, x_n$ are variables is called a $\boxed{\text{linear equation}}$. The real quantities $a_1, \dots, a_n$ are called $\boxed{\text{coefficients}}$, b is called a $\boxed{\text{constant}}$, while the variables $x_1, \dots, x_n$ are called *unknowns*.

Definition: Systems of linear equations A $\boxed{\text{system of linear equations}}$ or $\boxed{\text{linear system}}$ is a set of m linear equations each in n unknowns. A linear system has the form

$$\begin{array}{ccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & b_m \end{array}$$

where $a_{11}, \dots, a_{mn}$ and $b_1, \dots, b_m$ are constants.

A $\boxed{\text{solution}}$ to the linear system the referenced item is a sequence of n numbers $s_1, \dots, s_n$ so that *each* equation in the referenced item is satisfied when the substitutions $x_1 = s_1$, $x_2 = s_2, \dots, x_n = s_n$ are made.

- If a linear system has *no* solution, it is called $\boxed{\text{inconsistent}}$.
- If a linear system has at least one solution, it is called $\boxed{\text{consistent}}$. Note that an equation with infinitely many solutions is still consistent!

If two linear systems have exactly the same solutions, they are called $\boxed{\text{equivalent}}$.

Warning: Variable names We use x, y, z in geometric examples and $x_1, \dots, x_n$ for general systems.

Activity: Nonlinear equations

What are some equations that are not linear equations?

In Understanding a linear map: a preview, the question “can this output be produced?” became a question about whether a system can be solved.

Activity: Revisiting a requested output as equations

Return to Can this map produce the requested output?:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

Write $A\mathrm{x} = \mathbf{b}$ for the two targets

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \qquad \mathbf{b}_2 = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

What equations do these two targets produce? Which target is reachable?

Before introducing a general algorithm, we look at a few more small systems where the equations can be analyzed directly.

Activity: Forest animals

In a Wisconsin forest, there are robins and badgers. Together they have 18 heads and 56 legs. How many robins and badgers are in the forest?

Activity: Identifying constants and unknowns

Identify the constants and unknowns in each linear equation in Forest animals, i.e., the two linear equations $x + y = 18$ and $2x + 4y = 56$.

Activity: Another forest problem

In another Wisconsin forest, there are also robins and badgers. Together they have 18 heads, 36 eyes, and 56 legs. How many robins and badgers are in the forest? Can we use any new terminology to describe the resulting system of linear equations that we use to determine the number of robins and badgers?

Activity: Forest analysis

In a third Wisconsin forest, there are deer and badgers. Together they have 18 heads and 70 legs. How many deer and badgers are in the forest? Can we use any new terminology to describe this system?

Geometric examples

Recall that a line in the plane which passes through a point $P_0(x_0, y_0)$, and has slope m, can be written in *point-slope form*, as a set of points $P(x, y)$ which solve the equation

$$y - y_0 = m(x - x_0).$$

Activity: The geometric point of view

Consider the following three systems, inspired by our forest examples. Graph the lines defined by these equations and check how this reflects the number of solutions in each case.

Activity: The geometric point of view

Consider the following three systems, inspired by our forest examples. Graph the lines defined by these equations and check how this reflects the number of solutions in each case.
System 1

$$\begin{aligned} x + y &= 18 \\ 2x + 4y &= 56 \end{aligned}$$

Activity: The geometric point of view

Consider the following three systems, inspired by our forest examples. Graph the lines defined by these equations and check how this reflects the number of solutions in each case.
System 2

$$\begin{aligned} x + y &= 18 \\ 4x + 4y &= 72 \end{aligned}$$

Activity: The geometric point of view

Consider the following three systems, inspired by our forest examples. Graph the lines defined by these equations and check how this reflects the number of solutions in each case.
System 3

$$\begin{aligned} x + y &= 18 \\ 4x + 4y &= 56 \end{aligned}$$

The same word "solution" also has a geometric meaning. A solution set can be a point, a line, a plane, empty, or higher-dimensional.
To obtain an equation which describes a plane in three-dimensional space, we take a different approach to the point-slope form equation for lines in two-dimensional space. Instead of defining the equation in terms of the direction in which the plane travels, we instead describe the plane in terms of the one direction in which the plane *does not travel*, i.e., the direction at a right angle to the plane.

Definition: Normal vector equation for planes The plane passing through $P_0(x_0, y_0, z_0)$ with normal vector $\mathbf{n}$ is the set of all points $P(x, y, z)$ such that $\mathbf{x} - \mathbf{x}_0$ is orthogonal to $\mathbf{n}$. That is,

$$\mathbf{n} \cdot (\mathbf{x} - \mathbf{x}_0) = 0,$$

or in scalar form, if $\mathbf{n} = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}$ and $\mathbf{x}_0 = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}$,

$$n_1(x - x_0) + n_2(y - y_0) + n_3(z - z_0) = 0.$$

In general form, this equation is written as

$$ax + by + cz = d,$$

where $a = n_1$, $b = n_2$, $c = n_3$, and

$$d = n_1 x_0 + n_2 y_0 + n_3 z_0.$$

Activity

Find the equation of the plane passing through the point $P(1, 0, 3)$, normal to the vector $\mathbf{n} = \begin{bmatrix} 2 \\ 6 \\ 4 \end{bmatrix}$.

Definition: Parametric equation for a line The line through $P_0(x_0, y_0, z_0)$ with

$\boxed{\text{direction vector}}$ $\mathbf{d}$ is the set of points $P(x, y, z)$ of the form

$$\mathbf{x} = \mathbf{x}_0 + t\mathbf{d},$$

where $\mathbf{x}_0$ is the vector corresponding to P_0. In coordinates, if $\mathbf{d} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$, this says

$$(x, y, z) = (x_0 + td_1, y_0 + td_2, z_0 + td_3).$$

If d_1, d_2, d_3 are all nonzero, we can also write the line *in symmetric form* as the set of points $P(x, y, z)$ such that

$$\frac{x - x_0}{d_1} = \frac{y - y_0}{d_2} = \frac{z - z_0}{d_3}.$$

All three values in this equation are equal to the scalar t which occurs in the first equations.

Activity

Find a parametric equation for the line passing through the two points $(3, 1, 0)$ and $(-3, 1, 4)$.

Activity: A plane perpendicular to a line

Find the scalar equation of the plane through $P = (4, 2, 1)$ and perpendicular to the line

$$x = 1 + 5t, \qquad y = -1 + 4t, \qquad z = 2 + 3t.$$

Shapes in three-dimensional space described as the zeros of a single equation are in most cases *two-dimensional*, so lines must be described as points which satisfy *two* different equations. If a line in three-dimensional space is equal to the intersection of two planes passing through a point $\mathbf{x}_0$ (see textbook figure fig-1-2-intersection-of-two-planes), with normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, then the line is precisely the set of points $\mathbf{x}$ which satisfy the two equations

$$\mathbf{n}_1 \cdot (\mathbf{x} - \mathbf{x}_0) = 0$$

and

$$\mathbf{n}_2 \cdot (\mathbf{x} - \mathbf{x}_0) = 0.$$

Example: A line of intersection from two planes Find a parametric equation for the line of intersection of

$$x + y + z = 1, \qquad -x + 2y - 3z = -1.$$

First set $z = 0$. Then

$$x + y = 1, \qquad -x + 2y = -1.$$

Solving gives $x = 1$ and $y = 0$, so $P(1, 0, 0)$ is on the line.
Next set $z = 1$. Then

$$x + y = 0, \qquad -x + 2y = 2.$$

Solving gives $x = -\frac{2}{3}$ and $y = \frac{2}{3}$, so $Q(-\frac{2}{3}, \frac{2}{3}, 1)$ is also on the line.
A direction vector is

$$\mathbf{v} = \overrightarrow{PQ} = \begin{bmatrix} -\frac{5}{3} \\ \frac{2}{3} \\ 1 \end{bmatrix}.$$

Thus one parametrization is

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -\frac{5}{3} \\ \frac{2}{3} \\ 1 \end{bmatrix}.$$

This example shows how finding the intersection of planes leads naturally to solving systems of linear equations.

Warning: Parallel planes The method above assumes the two planes intersect in a line. If the normal vectors are parallel, then the planes are either identical or have no intersection.

Row operations

All of these examples have the same form. The coefficients form a matrix A, the unknowns form a vector $\mathbf{x}$, and the constants form a vector $\mathbf{b}$. The system is $A\mathbf{x} = \mathbf{b}$.

A linear system of m equations in n unknowns can be written as one matrix equation

$$A\mathbf{x} = \mathbf{b}.$$

The system is consistent exactly when $\mathbf{b}$ is a linear combination of the columns of A.

Activity: Matrix equation for a linear system

For

$$\begin{array}{rcrcrcr} x_1 & + & 2x_2 & + & 3x_3 & = & 6 \\ 2x_1 & - & 3x_2 & + & 2x_3 & = & 14 \\ 3x_1 & + & x_2 & - & x_3 & = & -2, \end{array}$$

write the system as $A\mathbf{x} = \mathbf{b}$.

Definition: Matrices corresponding to linear equations The coefficient matrix of a linear system of the form the referenced item is the $m \times n$ matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

The constant vector is

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

We can adjoin $\mathbf{b}$ to matrix A to create the $\boxed{\text{augmented matrix}}$ representing our linear system:

$$[A \mid \mathbf{b}] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right]$$

Note For the system $A\mathbf{x} = \mathbf{b}$, there are three related objects. The matrix A is the coefficient matrix. The vector $\mathbf{b}$ is the right-hand side. The augmented matrix is

$$[A \mid \mathbf{b}].$$

When solving a system, row-reduce the augmented matrix $[A \mid \mathbf{b}]$. Only columns to the left of the vertical line correspond to variables. The column to the right of the vertical line is not a variable column.

We will build a method of efficiently solving a large system of linear equations by applying one of several elementary operations to the system, which do not change the solutions to the system, with the hope of eventually simplifying the system to one in which the set of solutions is obvious.

Definition: Elementary operations The following operations, called $\boxed{\text{elementary operations}}$, can be performed on systems of linear equations, and produce $\boxed{\text{equivalent}}$ systems.

1. *Interchanging* the i-th and j-th equations.

2. *Multiplying* an equation by a nonzero constant.

3. *Replacing* the i-th equation by c times the j-th equation *plus* the i-th equation, where $i \neq j$, for some real quantity c.

We can track these operations more easily using an array/matrix to record the coefficients and constants.

Activity: Elementary operations with matrices

Consider the linear system

$$\begin{array}{ccccccc} & & y & - & z & = & 4 \\ 2x & & & + & 6z & = & 4 \\ 3x & + & 2y & + & 2z & = & -1 \end{array}$$

This system is reduced to a simpler system using elementary operations. Find the augmented matrix for the linear systems, and describe the operation that is performed to obtain this matrix from the previous one.

Notice that when we apply elementary operations to a system of linear equations, the corresponding rows of their corresponding matrices also change in an analogous way.

Definition: Elementary row operations An elementary row operation on a matrix M is any one of the following operations:

1. *Interchange* two rows.

2. *Multiply* a row by a nonzero scalar.

3. *Add* a multiple of one row to a different row.

Definition: Row equivalence Two matrices M and N of the same size are $\boxed{\text{row equivalent}}$ if N can be obtained from M by a finite sequence of elementary row operations.

Warning: Augmented matrices In the previous two definitions, the matrices M and N can be augmented matrices. In that case, row operations act on every column, including the right-hand-side column.

Theorem Consider two linear systems each of m equations in n unknowns. If the augmented matrices $[A \mid \mathbf{b}]$ and $[C \mid \mathbf{d}]$ are *row equivalent*, then the linear systems are $\boxed{\text{equivalent}}$, i.e., the systems have the same solutions.

The next step is to choose row operations systematically.

Gaussian elimination

Substitution works for small examples. Gaussian elimination is the systematic version: replace a system by an equivalent, simpler system.

Activity: Solving systems using back substitution

The following are augmented matrices (see Matrices corresponding to linear equations) representing systems of linear equations in three unknowns x, y, and z. Write the equations corresponding to each system of equations, then solve the systems.

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & -3 \end{array}\right]$$

Activity: Solving systems using back substitution

The following are augmented matrices (see Matrices corresponding to linear equations) representing systems of linear equations in three unknowns x, y, and z. Write the equations corresponding to each system of equations, then solve the systems.

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

Definition: Row echelon form and reduced row echelon form An $m \times n$ matrix M is in row echelon form (REF) if it satisfies the following properties:

1. All *zero rows*, if there are any, appear at the *bottom* of the matrix.
2. The *leftmost nonzero entry* in any nonzero row is a 1. This entry is called the leading one of its row.
3. For each nonzero row, the leading one appears to the *right and below* any leading ones in *preceding* rows.

We say M is in reduced row echelon form (RREF) if M is in REF and also satisfies:

- all columns containing leading ones contain no other non-zero entries.

A matrix in row echelon form appears as a staircase (or *echelon*) pattern of leading ones descending from the upper left corner of the matrix. Textbook figure fig-staircase gives a schematic of a matrix in row echelon form, where the asterisks denote arbitrary constants.

Activity: Identifying REF and RREF matrices

For each of the following matrices, determine whether it is in row echelon form, reduced row echelon form, both, or neither.

$$A = \begin{bmatrix} 1 & 3 & 0 & 0 & 1 \\ 0 & 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Activity: Identifying REF and RREF matrices

For each of the following matrices, determine whether it is in row echelon form, reduced row echelon form, both, or neither.

$$B = \begin{bmatrix} 1 & 4 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

Activity: Identifying REF and RREF matrices

For each of the following matrices, determine whether it is in row echelon form, reduced row echelon form, both, or neither.

$$C = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & 1 & -1 & 5 \\ 0 & 0 & 1 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Activity: Identifying REF and RREF matrices

For each of the following matrices, determine whether it is in row echelon form, reduced row echelon form, both, or neither.

$$D = \begin{bmatrix} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 0 & 1 & 6 \end{bmatrix}$$

Activity: Identifying REF and RREF matrices

For each of the following matrices, determine whether it is in row echelon form, reduced row echelon form, both, or neither.

$$E = \begin{bmatrix} 1 & 0 & 4 & 3 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & -5 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Definition: Pivot entries, positions, and columns For a matrix M in REF, each leading one is a pivot entry. Its location is a pivot position. A column containing a pivot position is a pivot column.

Note For an augmented matrix $[A \mid \mathbf{b}]$, only columns to the left of the vertical line are variable columns. A pivot to the right of the vertical line means the system is inconsistent.

Activity: Reading row reduction in code

The following code row reduces the augmented matrix for the two planes

$$x + y + z = 1, \qquad -x + 2y - 3z = -1.$$

```
import sympy as sp

M = sp.Matrix([
    [1, 1, 1, 1],
    [-1, 2, -3, -1],
])

M.rref()
```

Output:

```
(Matrix([
[1, 0,  5/3, 1],
[0, 1, -2/3, 0]]), (0, 1))
```

The output has the form

```
(reduced matrix, pivot columns)
```

SymPy numbers columns starting at 0. Thus `(0, 1)` means the first and second columns.
Read M as an augmented matrix $[A \mid \mathbf{b}]$: the first three columns are coefficient columns, and the last column is the right-hand side.

1. Which variable columns are pivot columns?
2. Which variable column is not a pivot column?
3. Let $z = t$. Use the reduced matrix to write x and y in terms of t.
4. Write the solution set in parametric vector form.
5. What kind of solution set is this?

Theorem Every matrix M is *row equivalent* to a *unique* matrix in RREF.

The Gaussian Algorithm gives a procedure to find a row-echelon matrix which is row equivalent to any given matrix. The Heuristic is to move from top to bottom and outside in until you obtain a matrix in row-echelon form.
Given a matrix, this algorithm will compute a matrix in row echelon form which is row-equivalent to the original matrix.

1. If the matrix consists entirely of zeros, stop — it is already in row-echelon form.

2. Otherwise, find the left-most column containing a nonzero entry a, and move the row containing that entry to the top position.

3. Now multiply the new top row by $1/a$ so that the row has a leading one.

4. By subtracting multiples of that row from rows below it, make each entry below this leading one equal to zero.

This completes the first row. Now Repeat Steps 1-4 on the matrix consisting of all other rows. The process stops when either no rows remain, or the remaining rows consist entirely of zeros.

Activity: Finding row echelon forms

Find a row echelon form of the given matrix A.

$$A = \begin{bmatrix} 0 & 0 & 1 & 2 \\ 3 & 3 & 6 & -9 \\ 2 & 3 & 0 & -2 \end{bmatrix}$$

Since we can solve systems of linear equations whose augmented matrices are in row echelon form using back substitution as in Solving systems using back substitution, The Gaussian Elimination Algorithm gives a practical method of solving any system of linear equations.

Activity: Solving a linear system using row echelon form

Solve the linear system

$$\begin{array}{rcrcrcr} x & + & 2y & + & 3z & = & 9 \\ 2x & - & y & + & z & = & 8 \\ 3x & & & - & z & = & 3 \end{array}$$

by using The Gaussian Elimination Algorithm to find a matrix in row echelon form which is row equivalent to the augmented matrix $[A \mid \mathbf{b}]$.

Note In Solving a linear system using row echelon form, every variable column has a pivot. Equivalently, the coefficient matrix can be row-reduced all the way to I_3. This is why the system has a unique solution.

Definition: Echelon Forms for Linear Systems of Equations If the augmented matrix of a linear system is a matrix in row echelon form, we say the linear system is in $\boxed{\text{echelon form}}$. The variables corresponding to variable columns with leading entries are called $\boxed{\text{leading variables}}$ (or $\boxed{\text{basic variables}}$), while the other variables are called $\boxed{\text{free variables}}$.

Back substitution allows us to write a general solution to a linear system by writing the basic variables in terms of the free variables.

Activity: Solving a system in REF

Let

$$[C \mid \mathbf{d}] = \left[\begin{array}{ccccc|c} 1 & 2 & 3 & 4 & 5 & 6 \\ 0 & 1 & 2 & 3 & -1 & 7 \\ 0 & 0 & 1 & 2 & 3 & 7 \\ 0 & 0 & 0 & 1 & 2 & 9 \end{array}\right]$$

be the augmented matrix of a linear system. Solve the system.

Activity: Analyzing an inconsistent system

Consider a linear system with augmented matrix

$$[C \mid \mathbf{d}] = \left[\begin{array}{cccc|c} 1 & 2 & -3 & 4 & 5 \\ 0 & 1 & 2 & 1 & -6 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Solve the system.

1. Write the augmented matrix $[A \mid \mathbf{b}]$.
2. Row-reduce $[A \mid \mathbf{b}]$ to a row echelon form $[C \mid \mathbf{d}]$.
3. If a row has the form

   $$[0\ 0\ \cdots\ 0 \mid c], \qquad c \neq 0,$$

   then the system is inconsistent.
4. Otherwise, pivot columns to the left of the vertical line give basic variables.
5. Non-pivot columns to the left of the vertical line give free variables.

6. Use back substitution to write the basic variables in terms of the free variables.

Row echelon matrices are not unique for a given starting matrix. For example,

$$\begin{bmatrix} 1 & 3 & 9 \\ 0 & 1 & -6 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 9 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

are row equivalent row echelon matrices, but they are not equal. However, the number of leading ones will be the same in each row equivalent row echelon matrix.

For now, rank is a row-reduction count: the number of pivots. Later we will reinterpret it as the number of independent directions transmitted by a matrix.

Definition: Rank of a matrix The rank of a matrix M is the number of pivots in any row echelon form row equivalent to M.

Activity: Finding the rank of a matrix

Find the rank of

$$C = \begin{bmatrix} 1 & 1 & 2 & 7 \\ 0 & 1 & 0 & 3 \\ 2 & 1 & 4 & 11 \end{bmatrix}$$

Activity: Finding the rank of a matrix (continued)

Find the rank of

$$D = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 2 & 1 & 2 & -1 \end{bmatrix}$$

Theorem Suppose $A\mathbf{x} = \mathbf{b}$ is a consistent system with n variables. Let r be the number of pivot columns to the left of the vertical line in a row echelon form of $[A \mid \mathbf{b}]$. Then the solution set is described using exactly $n - r$ parameters. In particular:

- if $r = n$, the system has a unique solution;
- if $r < n$, the system has infinitely many solutions.

The number r is the rank of the coefficient matrix A.

See the theorem. Together with the inconsistency test, it gives a method for finding the number of solutions to any given linear system.

Note: The Number of Solutions to a Linear System Any linear system belongs to exactly one of three categories.

- *No solution.* This occurs when a row echelon form of $[A \mid \mathrm{b}]$ has a row

$$[0\ 0\ \cdots\ 0 \mid c], \qquad c \neq 0.$$

- *A unique solution.* This occurs when the system is consistent and every variable column, meaning every column to the left of the vertical line, is a pivot column.
- *Infinitely many solutions.* This occurs when the system is consistent and at least one variable column, meaning a column to the left of the vertical line, is not a pivot column.

Note: Row-operation shorthand We often record row operations using the symbol R_i for row i. The notation $R_i \to R_i + cR_j$ means to replace row i by row i plus c times row j. All rows not named on the left side stay the same.

- $R_3 \to R_3 - 2R_1$ means replace row 3 by row 3 minus twice row 1.
- $R_2 \to \frac{1}{5}R_2$ means multiply row 2 by 1/5.
- $R_1 \leftrightarrow R_2$ means interchange rows 1 and 2.

Activity: Analyzing solution sets using rank

If

$$C = \left[\begin{array}{ccc|c} 1 & 1 & 2 & 7 \\ 0 & 1 & 0 & 3 \\ 2 & 1 & 4 & 11 \end{array}\right]$$

represents the augmented matrix for a system of 3 equations and 3 variables, what can we say about the solution set of the system?

Activity: Analyzing solution sets using rank (continued)

If

$$D = \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 2 & 1 & 2 & -1 \end{array}\right]$$

represents the augmented matrix for a system of 3 equations and 3 variables, what can we say about the solution set of the system?

Homogeneous systems and basic solutions

The preview also asked what changes a map forgets. That is the special system $A\mathrm{x} = \mathbf{0}$.

Definition: Homogeneous systems If a linear system is of the form

$$\begin{array}{ccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & 0 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & 0 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & 0 \end{array}$$

it is called a $\boxed{\text{homogeneous system}}$.

- Note that $x_1 = x_2 = \cdots = x_n = 0$ is always a solution to a homogeneous system. It is called the $\boxed{\text{trivial}}$ solution.
- A solution to a homogeneous system where *not* all x_i are 0 is called a $\boxed{\text{nontrivial}}$ solution.

Activity: Solving a homogeneous system

Solve the homogeneous system

$$\begin{aligned} 2x_1 + 4x_3 + 6x_4 &= 0 \\ x_1 + 2x_2 + 7x_4 &= 0 \end{aligned}$$

Activity: A plane through three points without cross products

Find an equation for the plane containing

$$P(1,1,-2), \qquad Q(0,2,1), \qquad R(-1,-1,0),$$

(*without* using the cross-product, if you happen to know what that is).

Note: Normal vectors from homogeneous systems The plane-through-three-points activity found a normal vector by solving a homogeneous system. Basic solutions below make this procedure systematic.

Note: Existence of solutions to homogeneous equations A homogeneous system of m linear equations in n unknowns *always* has a nontrivial solution if $n > m$, that is if the number of unknowns exceeds the number of equations.

Activity: Thinking about homogeneous systems

Does a homogeneous system *need* to have more variables than equations in order to have infinitely many solutions?

The solution set of a homogeneous system can be written in vector form.

Activity: Solutions as linear combinations

Rewrite the general solution to Solving a homogeneous system as an expression involving a linear combination of certain column vectors.

Fact If we consider solutions to a homogeneous system as vectors, then any linear combination of solutions to a homogeneous system is again a solution to the homogeneous system.

Activity: Finding basic solutions

Consider the homogeneous system whose augmented matrix, when put in RREF, is

$$[A \mid \mathbf{0}] = \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Solve the system, and write the solutions as a linear combination of particular solutions.

Definition: Basic solutions to homogeneous linear equations The Gaussian elimination algorithm produces, for each free variable, a solution to the homogeneous equation $A\mathbf{x} = \mathbf{0}$, by setting that free variable equal to one, and all other free variables equal to 0. These solutions are called

$\boxed{\text{basic solutions}}$, one for every parameter. Every solution can be written uniquely as a linear combination of the basic solutions.

Theorem Let A be a rank r matrix with m rows and n columns. Consider the homogeneous system in n variables with A as the coefficient matrix. Then:

1. The system has exactly $n - r$ basic solutions, one for each parameter. Note that $n - r$ is also the number of *free* variables of the linear system when the augmented matrix $[A \mid \mathbf{0}]$ is reduced to row echelon form (recall Echelon Forms for Linear Systems of Equations).
2. Every solution is a linear combination of these basic solutions.

For a homogeneous system, row-reducing $[A \mid \mathbf{0}]$ is equivalent to row-reducing A while remembering that the right-hand side stays zero. The next section gives the solution set of $A\mathbf{x} = \mathbf{0}$ a name, and from that point on we work with A alone.
Homogeneous solution sets are closed under linear combinations. Subspaces gives a name to sets with this behavior: subspaces.